Q07: Comprehension Check

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October 2026

Comprehension check. A lab has a fixed compute budget and plans to spend it all on a very large model trained on a relatively small dataset. Citing the compute-optimal result, what is likely wrong with this plan, and what does $C \approx 6ND$ tell them to do instead?

Answer. The plan over-parameterizes and under-trains: Chinchilla showed that for a fixed compute budget, loss is minimized by scaling model size N and training tokens D *together* (roughly equally), not by maximizing N alone. Since $C \approx 6ND$ is fixed, pouring budget into N forces D down, leaving the big model undertrained and achieving *higher* loss than a smaller model trained on more tokens at the same cost. They should rebalance toward a smaller N and a larger D to hit the compute-optimal point.